

AP Statistics FRQ Practice Exam 10

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APSTAT-FRQ10-001

Section: FRQ

Type: frq

Primary Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Secondary Unit: Unit 4, Inference for Quantitative Data: Means

FRQ Type: Formulate Questions and Collect Data

Prompt:

A harbor authority wants to compare a standard electronic chart with a new augmented route overlay for helping harbor pilots identify a safe route through an unfamiliar channel. From its 180 licensed harbor pilots, the authority takes a simple random sample of 40 pilots. Each selected pilot will complete two comparable simulator scenarios, using the standard chart in one scenario and the augmented overlay in the other. The response variable is the number of seconds needed to identify the correct route.

The design team initially proposes that every pilot use the standard chart first in Scenario A and the augmented overlay second in Scenario B. Before data are collected, the design can be revised. Assume that a properly randomized crossover design is ultimately used. The observed mean times are summarized below.

Display	Mean time
Standard chart	74.6 seconds
Augmented overlay	66.1 seconds

For each pilot, define d = overlay time – standard-chart time. The sample mean of the 40 paired differences is -8.5 seconds.

- Identify the population, sample, experimental units, and response variable.
- State a statistical question that the study can answer, and define an appropriate population parameter for that question.
- Explain why the initial fixed-order proposal is flawed. Describe a randomized matched-pairs design that addresses both scenario and order effects.
- Calculate and interpret the observed mean paired difference.
- Explain what roles random sampling and random assignment play in the scope of the study's conclusions. Include one remaining limitation of the simulator study.

Scoring Guide:

Total Points: 10

(a) 2 points

- Point 1: Identifies the population as all 180 licensed harbor pilots and the sample as the 40 pilots selected for the study.
- Point 2: Identifies each selected pilot as an experimental unit and route-identification time in seconds as the quantitative response variable.

(b) 2 points

- Point 1: States a comparative statistical question about whether the two displays differ in mean route-identification time for the authority's pilots under comparable simulator conditions.
- Point 2: Defines μ_d as the population mean of overlay time minus standard-chart time for all 180 pilots under the study conditions.

(c) 3 points

- Point 1: Explains that the fixed proposal confounds display type with scenario and order, so practice, fatigue, or scenario difficulty could create an apparent display effect.
- Point 2: Uses each pilot as a matched pair, with every pilot using each display exactly once on the two comparable scenarios.
- Point 3: Describes valid randomization or counterbalancing, such as randomly assigning pilots equally among the four combinations of display-to-scenario assignment and display order, while keeping other simulator conditions fixed.

(d) 1 point

- Point 1: Calculates $66.1 - 74.6 = -8.5$ seconds and interprets this as the augmented overlay being faster by 8.5 seconds on average for the sampled pilots.

(e) 2 points

- Point 1: Explains that random assignment of displays within pilots supports a cause-and-effect conclusion for the sampled pilots, provided important carryover effects are controlled.
- Point 2: Explains that the simple random sample supports generalization to the 180 pilots, but conclusions may not extend to real emergencies because simulator behavior or residual learning and fatigue may differ from field conditions.

Model Guidance:

The experimental units are the pilots, not the two observations from each pilot. Because every pilot tries both displays, the natural comparison is the within-pilot difference $d = \text{overlay} - \text{standard}$. Assigning every standard-chart trial first would make display inseparable from order and scenario. A sound crossover plan balances both which display is used in each scenario and which display is used first.

The mean difference is -8.5 seconds, so the overlay was faster on average in the sample. Random assignment addresses causal comparison of the displays, whereas the simple random sample addresses representativeness of the authority's pilots. Neither feature guarantees that results from a simulator reproduce behavior during actual harbor emergencies.

Expected Points:

10

APSTAT-FRQ10-002

Section: FRQ

Type: frq

Primary Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Secondary Unit: None

FRQ Type: Analyze Data, One- or Two-Variable Data and Probability

Prompt:

A wildlife agency installs tracking tags in one of two terrain settings. Historical installation records give the following probabilities for terrain and for receiving a usable signal throughout the tag's first week.

Terrain setting	$P(\text{setting})$	$P(\text{usable signal} \mid \text{setting})$
Open grassland	0.65	0.92
Dense woodland	0.35	0.74

Let S denote the event that a newly installed tag provides a usable first-week signal.

(a) Sketch a probability tree for terrain setting and first-week signal status. Label all branch probabilities and the four joint probabilities at the endpoints.

(b) Calculate $P(S)$. Interpret the probability in context.

(c) Given that a tag provides a usable signal, calculate the probability that it was installed in dense woodland. Determine whether terrain setting and usable-signal status are independent, and justify your answer.

For one project, 12 tags will be installed independently. Each installation has the same terrain mix and conditional signal probabilities shown in the table. Let Y be the number of the 12 tags that provide usable signals.

(d) Identify the probability distribution of Y and verify its conditions. Calculate $P(Y \geq 11)$, and calculate and interpret the mean and standard deviation of Y .

Scoring Guide:

Total Points: 10

(a) 2 points

- Point 1: Draws or clearly describes a two-stage tree with first branches 0.65 for open grassland and 0.35 for dense woodland, followed by usable/not-usable branches 0.92/0.08 and 0.74/0.26, respectively.
- Point 2: Labels the endpoint joint probabilities as 0.598, 0.052, 0.259, and 0.091 for open-and-usable, open-and-not-usable, woodland-and-usable, and woodland-and-not-usable.

(b) 2 points

- Point 1: Calculates $P(S) = (0.65)(0.92) + (0.35)(0.74) = 0.857$.
- Point 2: Interprets 0.857 as the long-run proportion, about 85.7%, of installations with this terrain mix that would provide a usable first-week signal.

(c) 2 points

- Point 1: Calculates $P(\text{woodland} | S) = 0.259/0.857 \approx 0.302$.
- Point 2: Concludes that the events are not independent because $P(S | \text{woodland}) = 0.74$ differs from $P(S) = 0.857$, or gives an equivalent valid comparison.

(d) 4 points

- Point 1: Identifies Y as $\text{Binomial}(12, 0.857)$ and verifies a fixed 12 installations, two signal outcomes, independent installations, and constant success probability 0.857.
- Point 2: Calculates $P(Y \geq 11) = C(12, 11)(0.857)^{11}(0.143) + (0.857)^{12} \approx 0.4712$.
- Point 3: Calculates $\mu_Y = np = 12(0.857) = 10.284$ usable tags and interprets it as the long-run average count per group of 12 installations.
- Point 4: Calculates $\sigma_Y = \sqrt{np(1 - p)} = \sqrt{12(0.857)(0.143)} \approx 1.213$ tags and interprets it as the typical distance of Y from its mean over many groups of 12.

Model Guidance:

The tree separates the terrain mix from the terrain-specific signal rates. Multiplying along a path gives a joint probability, and adding the two usable-signal paths gives $P(S) = 0.857$. For the reverse conditional probability, the woodland-and-usable path is divided by the total usable-signal probability. The different conditional success rates also show that terrain and signal status are associated.

For the 12-tag project, the stated independence and common installation mechanism make the count binomial with $p = 0.857$. The upper-tail calculation must include both 11 and 12 usable tags. A mean of 10.284 is a long-run average, not a claim that a particular project can produce a fractional tag. A probability histogram for Y would place bars at integers 0 through 12, with most mass near 10 through 12 and a longer tail toward smaller counts.

Expected Points:

10

APSTAT-FRQ10-003

Section: FRQ

Type: frq

Primary Unit: Unit 4, Inference for Quantitative Data: Means

Secondary Unit: Unit 1, Exploring One-Variable Data and Collecting Data

FRQ Type: Analyze Data, Statistical Inference (Confidence Interval or Significance Test)

Prompt:

A supplier produces a lot of 1,000 spools of biodegradable garden twine. A quality engineer takes a simple random sample of 25 spools and measures the breaking force of one standardized twine segment from each sampled spool. Breaking force is measured in newtons. The sample results are

$n = 25$

$\bar{x} = 118.6$ newtons

$s = 12.5$ newtons

A dotplot of the 25 measurements is roughly symmetric with no apparent outliers. For 24 degrees of freedom, use $t^* = 2.064$ for a 95% confidence interval.

- (a) Define the population parameter of interest.
- (b) Identify the appropriate confidence-interval procedure and check all conditions for using it.
- (c) Calculate the standard error and margin of error, and construct a 95% confidence interval for the population mean breaking force.
- (d) Interpret the confidence interval in context.
- (e) The supplier claims that the lot's true mean breaking force is at least 115 newtons. Evaluate whether the interval provides convincing support for that claim. Explain why the sample mean alone is not sufficient.

Scoring Guide:

Total Points: 10

(a) 1 point

- Point 1: Defines μ as the true mean breaking force, in newtons, of standardized twine segments from all 1,000 spools in the production lot.

(b) 3 points

- Point 1: Selects a one-sample t confidence interval for a population mean because the response is quantitative and the population standard deviation is unknown.
- Point 2: Cites the simple random sample and verifies the 10% condition because $25 \leq 0.10(1,000) = 100$, supporting independence.
- Point 3: Uses the roughly symmetric, outlier-free dotplot to justify the t procedure for the sample of 25.

(c) 3 points

- Point 1: Calculates $SE = s/\sqrt{n} = 12.5/\sqrt{25} = 2.5$ newtons.
- Point 2: Calculates the margin of error as $2.064(2.5) = 5.16$ newtons.
- Point 3: Constructs 118.6 ± 5.16 , giving the interval (113.44, 123.76) newtons.

(d) 1 point

- Point 1: States that the engineer is 95% confident that the true mean breaking force for the lot of 1,000 spools is between 113.44 and 123.76 newtons.

(e) 2 points

- Point 1: Concludes that the interval does not provide convincing 95%-confidence support for a mean of at least 115 newtons because plausible values in the interval are below 115.
- Point 2: Explains that $\bar{x} = 118.6$ is subject to sampling variability, so a point estimate above 115 does not by itself establish that the population mean is at least 115.

Model Guidance:

This is a one-sample t interval because the goal is a mean and only the sample standard deviation is known. The random sample, small sampling fraction, and approximately symmetric distribution without outliers support the procedure. The standard error is 2.5 newtons, so the 95% margin of error is 5.16

and the interval is (113.44, 123.76).

The interval refers to the population mean, not to 95% of individual spools. Although the sample mean exceeds 115, the interval includes possible mean values below 115. Therefore, the data do not justify the supplier's one-sided "at least 115" claim at the confidence level used.

Expected Points:

10

APSTAT-FRQ10-004

Section: FRQ

Type: frq

Primary Unit: Unit 5, Regression Analysis

Secondary Unit: Unit 2, Probability, Random Variables, and Probability Distributions

FRQ Type: Interpret Results, Integrated Analysis and Justification

Prompt:

A regional food-bank sorting center takes a simple random sample of 80 one-hour periods from all weekday daytime operating periods during the most recent three months. For each sampled hour, the center records

x = number of unsorted pallets waiting at the start of the hour

y = median driver check-in wait during the hour, in minutes.

The observed x -values range from 1 to 9. A scatterplot shows a moderately strong, positive, roughly linear relationship with no apparent unusual observations. Least-squares output is shown below.

Term	Estimate
Constant	3.200
Unsorted pallets	1.800

$r = 0.872$

$r^2 = 0.760$

$s = 2.400$ minutes

For decision modeling within the observed x -range, suppose the residual wait at each integer value of x is approximately Normal with mean 0 and standard deviation 2.4 minutes. One sampled hour had $x = 6$ pallets and $y = 17.5$ minutes.

(a) Interpret the slope and r^2 in context.

(b) Calculate the predicted wait and residual for the sampled hour with $x = 6$ and $y = 17.5$. Interpret the sign of the residual.

(c) Under the Normal residual model, calculate the probability that an hour beginning with 6 unsorted pallets has a median driver wait greater than 18 minutes. Interpret the result.

(d) The center will open an auxiliary sorting line whenever the modeled probability of a median wait greater than 18 minutes exceeds 0.10. Determine the smallest integer number of starting pallets that triggers the rule. Justify your answer quantitatively. You may use $z_{0.90} = 1.2816$.

(e) An operations manager says, "Reducing the starting backlog from 7 pallets to 5 pallets will cause every hour's median wait to decrease by exactly 3.6 minutes." Evaluate the claim using the model, the study design, and the scope of the data.

Scoring Guide:

Total Points: 10

(a) 2 points

- Point 1: Interprets the slope: for each additional unsorted pallet at the start of an hour, the model predicts an increase of 1.8 minutes in the hour's median driver wait, on average.
- Point 2: Interprets $r^2 = 0.760$ as 76.0% of the variation in median driver wait among the sampled hours being explained by the linear relationship with starting pallet count.

(b) 2 points

- Point 1: Calculates $\hat{y} = 3.2 + 1.8(6) = 14.0$ minutes and the residual $17.5 - 14.0 = 3.5$ minutes.
- Point 2: Interprets the positive residual to mean that the observed median wait was 3.5 minutes longer than the model predicted for an hour starting with 6 pallets.

(c) 2 points

- Point 1: Standardizes the threshold using $z = (18 - 14.0)/2.4 \approx 1.67$ and calculates $P(Y > 18 \mid x = 6) = P(Z > 1.67) \approx 0.0478$.
- Point 2: Interprets the result as a modeled probability of about 4.8% that an hour starting with 6 pallets has a median driver wait above 18 minutes.

(d) 2 points

- Point 1: Sets the 10% upper-tail boundary using $18 - (3.2 + 1.8x) = 1.2816(2.4)$, obtaining $x \approx 6.51$ as the transition point.
- Point 2: Selects 7 pallets as the smallest integer trigger and verifies that the modeled upper-tail probability is about 0.180 at $x = 7$, whereas it is about 0.048 at $x = 6$.

(e) 2 points

- Point 1: Explains that the fitted line predicts a $1.8(7 - 5) = 3.6$ minute difference in mean response, not an exact decrease for every hour, because individual residuals vary.
- Point 2: Rejects the causal wording because backlog was observed rather than randomly imposed and notes that generalization is limited to comparable weekday daytime periods represented by the three-month sampling frame; possible operational confounders could affect both backlog and wait.

Model Guidance:

The fitted equation is $\hat{y} = 3.2 + 1.8x$. At six pallets it predicts 14.0 minutes, so the observed 17.5-minute wait has a positive 3.5-minute residual. The residual model treats actual waits as varying around the fitted mean. Thus, at $x = 6$ the upper-tail probability beyond 18 minutes is about 0.0478.

For the decision threshold, an upper-tail probability above 0.10 occurs when the standardized cutoff is below 1.2816. Solving gives $x >$ about 6.51, so the first integer trigger is 7. This decision is model-based and remains within the observed range. The slope supports a predicted average difference of 3.6 minutes between $x = 7$ and $x = 5$, but residual variation prevents an every-hour guarantee. Because the study is observational, it also cannot establish that actively removing two pallets would itself cause the predicted change.

Expected Points:

10

SERIES DO-NOT-REPEAT BANK UPDATE, APSTAT-FRQ10

- APSTAT-FRQ10-001: Unit 1 primary; Unit 4 secondary; SRS of 40 of 180 harbor pilots in a randomized crossover simulator experiment comparing standard charts with augmented route overlays, with route-identification time and paired difference overlay minus standard; diagnose fixed order/scenario confounding, counterbalance display-to-scenario assignment and order, interpret mean difference -8.5 seconds, and separate causal inference, population generalization, and simulator limitations; targets confusing observations with experimental units and overlooking carryover or order effects.
- APSTAT-FRQ10-002: Unit 2 primary; terrain/signal probability tree with open-grassland probability 0.65 and conditional success 0.92 versus dense-woodland probability 0.35 and conditional success 0.74; calculate joint endpoints, $P(\text{signal}) = 0.857$, reverse conditional $P(\text{woodland} \mid \text{signal}) \approx 0.302$, dependence, and $\text{Binomial}(12, 0.857)$ results $P(Y \geq 11) \approx 0.4712$, $\mu = 10.284$, $\sigma \approx 1.213$; targets reversing conditional probabilities and omitting a binomial upper-tail outcome.
- APSTAT-FRQ10-003: Unit 4 primary; Unit 1 secondary; one-sample t confidence interval for mean standardized breaking force in a lot of 1,000 biodegradable-twine spools using an SRS of 25, $\bar{x} = 118.6$ N, $s = 12.5$ N, symmetric outlier-free dotplot, and $t^* = 2.064$; obtain SE 2.5, margin 5.16, and 95% interval (113.44, 123.76), then reject support for an at-least-115-N claim; targets interpreting an interval as a range for individuals and treating a point estimate as proof of a threshold claim.
- APSTAT-FRQ10-004: Unit 5 primary; Unit 2 secondary; SRS of 80 weekday daytime food-bank operating hours with starting unsorted pallets and median driver wait, regression $\hat{y} = 3.2 + 1.8x$, $r = 0.872$, $r^2 = 0.760$, residual SD 2.4 minutes, and Normal residual model; calculate prediction/residual at $x = 6$, upper-tail wait probability, derive a 7-pallet decision trigger for probability above 0.10, and distinguish predicted average association from exact or causal change; targets using r^2 as a percentage of observations, treating a regression prediction as guaranteed, and making causal claims from observational data.